UNIT 2 TEST A: WRITTEN

UNIT 2 TEST · DAY 2 · WRITTEN · VERSION A · TEACHER KEY

NAME DATE PERIOD SCORE / 14

You may use a periodic table. Write every configuration in order of filling. In the orbital diagram, each electron gets one arrow.

Part 1 Longhand configurations

8 points

Number 1 also needs its orbital diagram and the number of half-filled orbitals.

1 — 4 pts

15 **P** Phosphorus 30.97

Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ **Half-filled orbitals** 3

Orbital diagram



2 — 2 pts

23 **V** Vanadium 50.94

Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^3$

3 — 2 pts

31 **Ga** Gallium 69.72

Configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^1$

1 is 4 pts, configuration 2, diagram 1, half-filled count 1. 2 and 3 are 2 pts each, 1 off for one wrong sublevel.

Part 2 Noble-gas shorthand

6 points

4

48 **Cd** Cadmium 112.41

$[Kr]\ 5s^2\ 4d^{10}$

5

80 **Hg** Mercury 200.59

$[Xe]\ 6s^2\ 4f^{14}\ 5d^{10}$

6

24 **Cr** Chromium 52.00

$[Ar]\ 4s^1\ 3d^5$

2 pts each, noble-gas core 1 and the rest in order 1. Chromium is the exception, so the predicted $[Ar]\ 4s^2\ 3d^4$ earns 1.